

4-minute presentation | max 10 slides

Detecting Depression Risk from Social Media Text

An interpretable machine learning triage workflow for Reddit mental health posts

Core claim: prioritise posts for trained human review - not automatic clinical diagnosis.

Why this problem matters

Social media can surface distress signals, but the use case must stay cautious.

5.7%

estimated adults affected by
depression globally

WHO cited in report

34,282

cleaned Reddit posts in the R
demo

after cleaning

2 classes

depression-related vs ADHD
comparison posts

more realistic than neutral text

Opportunity

Public mental health communities contain long-form posts about symptoms, medication, stigma, relationships and daily functioning. These posts may reveal early support needs.

Risk

Platform text is noisy and sensitive. Subreddit labels are not clinical diagnoses, and misclassification can create harm or extra workload.

Project framing

The project narrows the goal to post-level triage:

- score public posts
- explain influential terms
- route high-risk or uncertain cases to trained reviewers
- avoid identity inference and automatic diagnosis

Report evidence: Introduction and Abstract identify depression as a public health challenge and define the system as decision support, not diagnosis.

Research gap and project positioning

The novelty is in the workflow and governance, not just another classifier.

Common limitations in existing approaches

- performance-first classifiers can be hard to trust
- platform or self-disclosure labels are often treated too much like clinical labels
- false negatives and false positives have different costs
- many studies stop at model metrics without an operational review workflow

Shift

This project's contribution

- ADHD posts as a more challenging comparison group
- Logistic Regression as the main interpretable baseline
- coefficient-level explanations for human reviewers
- threshold-aware evaluation using Accuracy, Precision, Recall, F1 and AUC
- privacy-safe reporting and human oversight

Rubric alignment: clear problem, significance, business value, novelty and flow.

Data source and preprocessing

The analysis.Rmd turns raw Reddit CSV files into a reproducible text-classification dataset.

From code to dataset

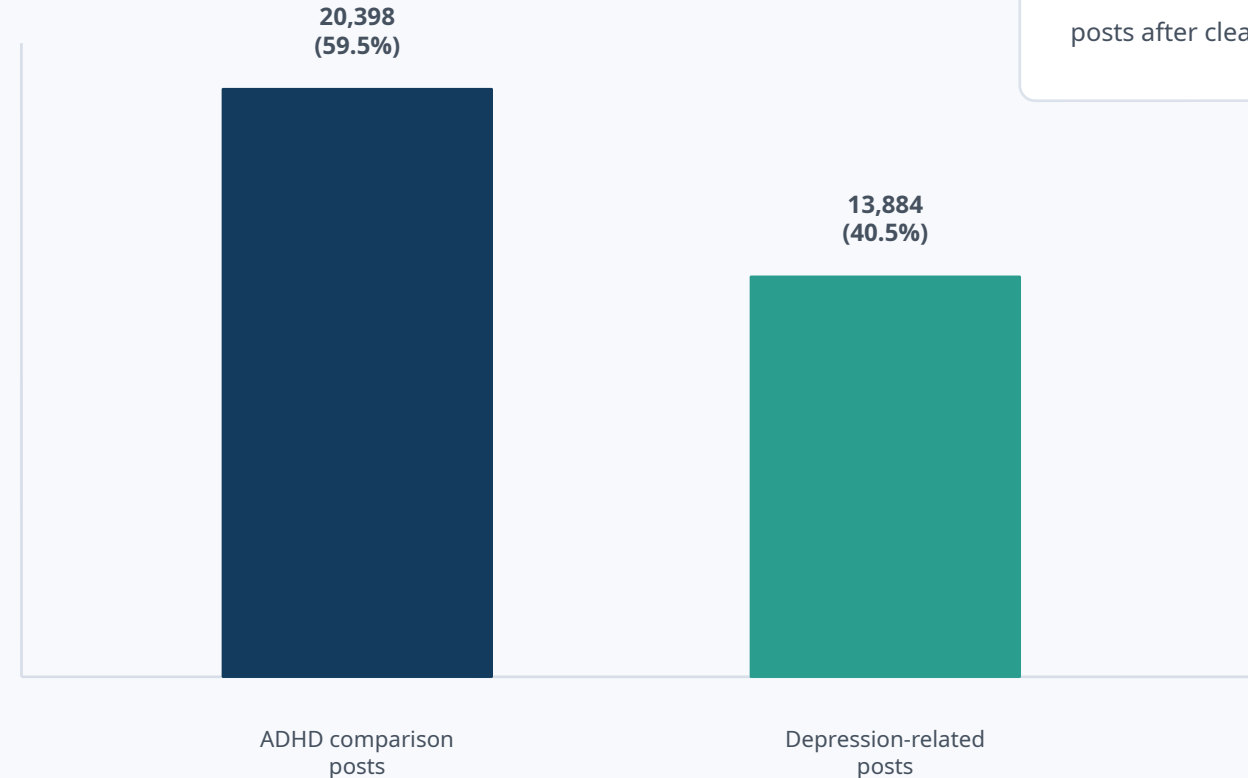
Input files:

- depression.csv from r/depression
- adhd.csv from r/ADHD

Cleaning logic:

- combine title + body
- remove [removed] and [deleted]
- remove empty or missing text
- create binary label
- save privacy-safer metadata version

Cleaned class distribution



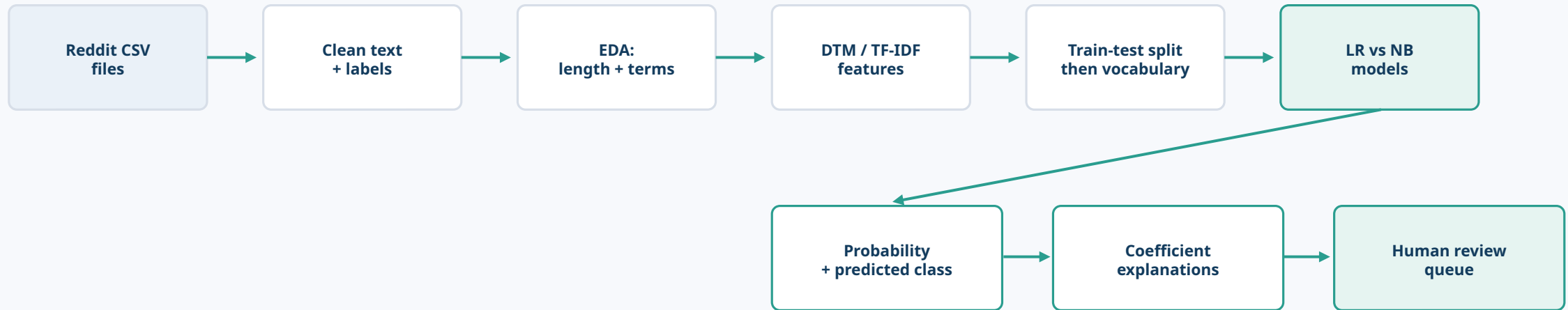
34,282

posts after cleaning

Moderately imbalanced, but usable for supervised learning when precision, recall and F1 are also reported.

Method and analysis pipeline

The R demonstration keeps the pipeline simple, explainable and auditable.



Evaluation logic

Accuracy alone is not enough. The report evaluates Accuracy, Precision, Recall, F1, AUC and error distribution because false negatives and false positives have different costs in triage.

Rmd core model code

```
logistic_model<-glm(label~.,  
  data = train_matrix, family = binomial)  
  
nb_model<-naiveBayes(x = train_x,y = train_y)
```

Results: interpretable model performs strongly

Logistic Regression balances discrimination and explainability.

0.9024

Logistic Regression accuracy

test set

0.9577

Logistic Regression AUC

ROC-AUC

0.8248

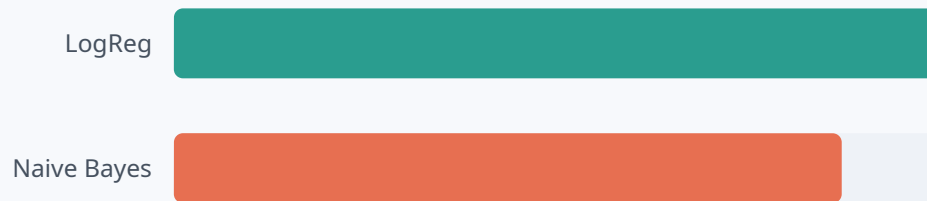
Naive Bayes AUC

baseline

Interpretation

The result does not prove clinical diagnosis. It shows that an interpretable baseline can separate depression-related posts from ADHD comparison posts well enough for a cautious triage demonstration.

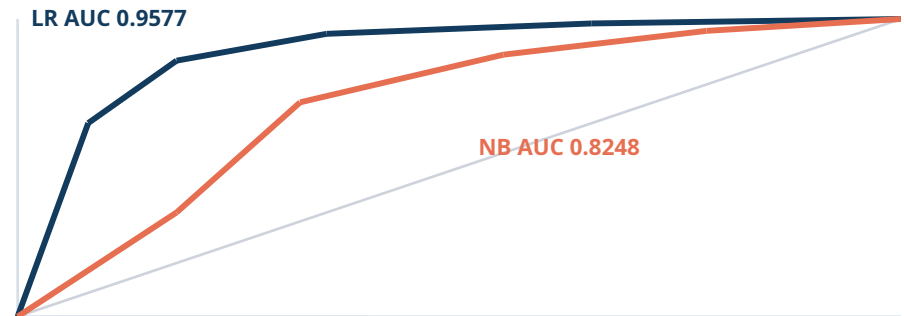
AUC comparison



Main message

The preferred model is not 'perfect': it is strong, transparent and easier to audit.

ROC intuition: curve closer to top-left = better discrimination



Report results section: LR accuracy = 0.9024; LR AUC = 0.9577; NB AUC = 0.8248.

Interpretability and error costs

A high score still requires human review because both false negatives and false positives matter.

How to read coefficients

Logistic Regression coefficients indicate which stemmed terms push a post toward the depression-related class or the ADHD comparison class.

They are model explanations, not psychological theory. Some terms may reflect subreddit norms, medication names or community language.

Logistic Regression error summary



Why human review? Even a strong classifier can make errors; outputs become review priorities, not final decisions.

False negative risk

A depression-related post may be missed; this matters for early support.

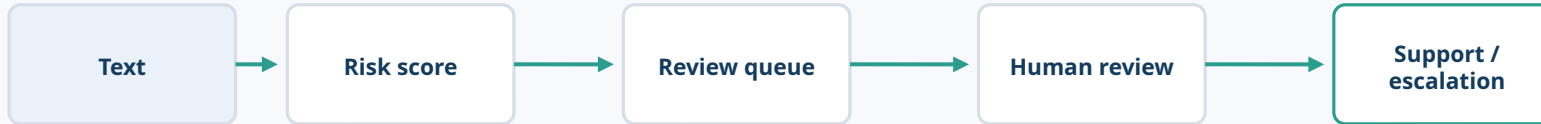
False positive risk

Reviewers may spend time on lower-priority posts or label users unfairly.

Governance, value and next steps

The deployable contribution is a transparent triage workflow with safeguards.

Value chain



Governance safeguards

- least-privilege access
- restrict raw text access
- pseudonymous IDs
- versioned data cleaning and model artifacts
- audit logs and retention rules
- no punitive moderation

Next steps

- consent-based or ethically sourced data
- clinical/scale-based labels
- threshold calibration with reviewers
- subgroup fairness checks
- model card and monitoring

Take-away

Interpretable social media text analysis is feasible for early-risk triage, but the responsible product is not an automatic diagnosis tool. It is a governed decision-support workflow.

Final message: help support teams see risky posts earlier - while keeping humans, privacy and accountability in the loop.

Report governance section and conclusion; specification requires a concise 4-minute presentation under 10 slides.